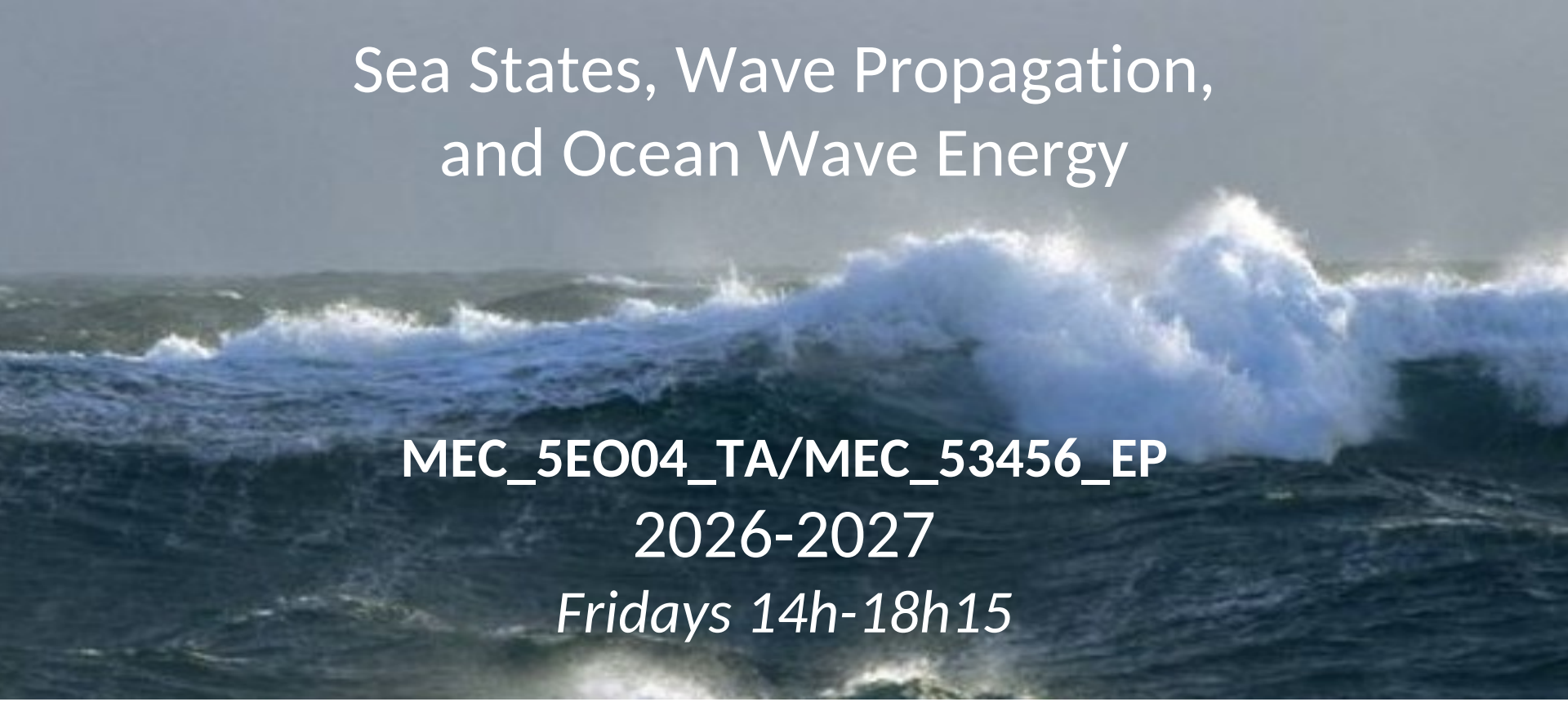


Sea States, Wave Propagation, and Ocean Wave Energy

MEC_5EO04_TA/MEC_53456_EP

2026-2027

Fridays 14h-18h15

Jeffrey Harris
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Marissa Yates
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ÉCOLE NATIONALE DES PONTS ET CHAUSSÉES

INSTITUT POLYTECHNIQUE DE PARIS

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Why study waves?















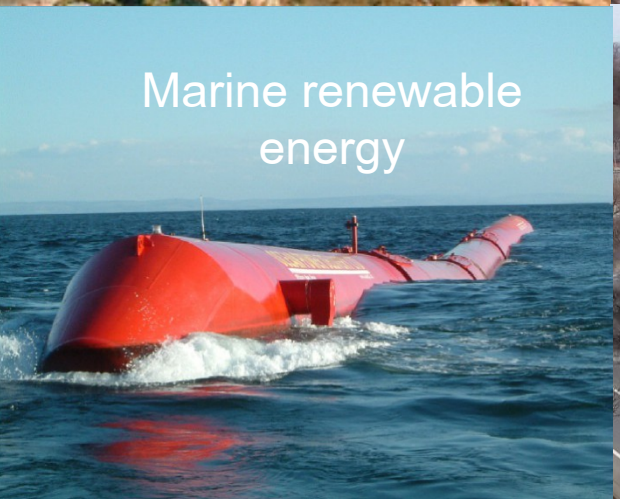




Why study waves?







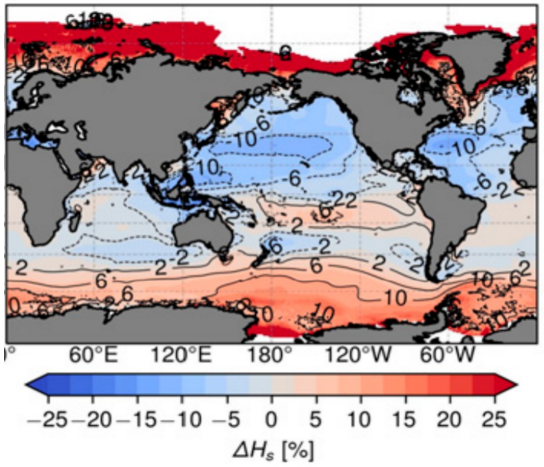


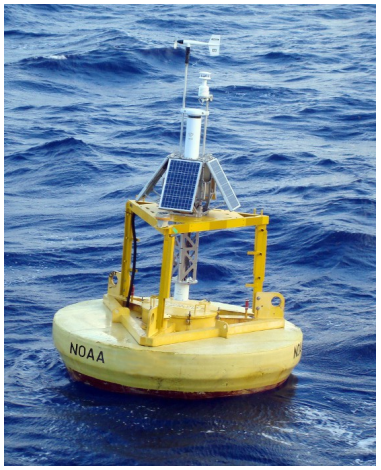


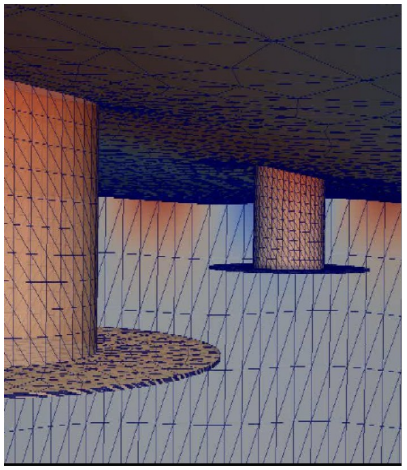
Course objectives

Prepare students to work in the fields of wave dynamics, coastal engineering, ocean wave energy, …









by learning to:

- describe wave characteristics,
- understand the physical processes governing wave transformation,
- choose an appropriate numerical modeling approach for a given application,
- understand the physical processes governing wave-body interactions,and
- apply an appropriate model to simulate wave-structure interactions.

(Meucci et al., 2024, Fig. 17 ; https://news.sky.com/story/worlds-largest-offshore-wind-farm-opens-off-cumbria-coast-11491285)

Course organization

I. Wave dynamics
II. Numerical modeling
III. Wave-structure interactions

Structure:

Lectures
In-class exercises
Project (via TD's)
Mini-presentations

Grading:

Class participation – 20%
Mini-presentations – 20%
Project reports – 30%
Final project presentation – 30%

Course schedule

I. Wave dynamics	Project	Lecturer	Date
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	25/09
2. Linear wave theory	Characterizing offshore waves	M. Yates	09/10
3. Nearshore wave propagation	Calculating extreme wave conditions	M. Yates	17/10
4. Coastal hydrodynamics	**Student mini-presentations**	M. Yates	**06/11****
II. Numerical wave modeling			
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	**23/10****
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	13/11
III. Wave-structure interactions			
7. Dynamics of a body in waves	Wave load estimation	J. Harris	20/11
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	04/12
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	11/12
10. Final project presentations 14h - 18h15			***18/12***

I. Wave dynamics

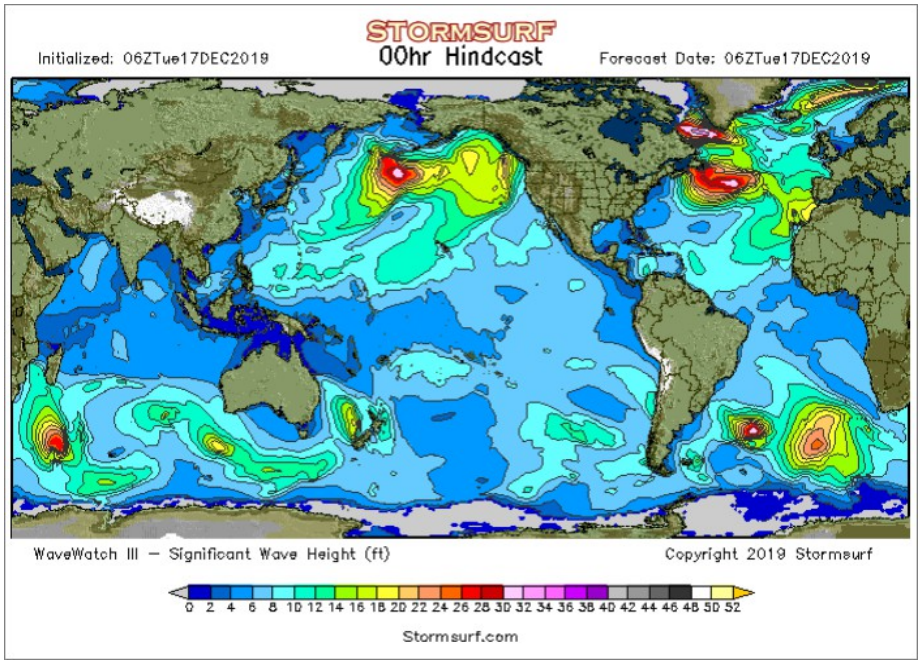


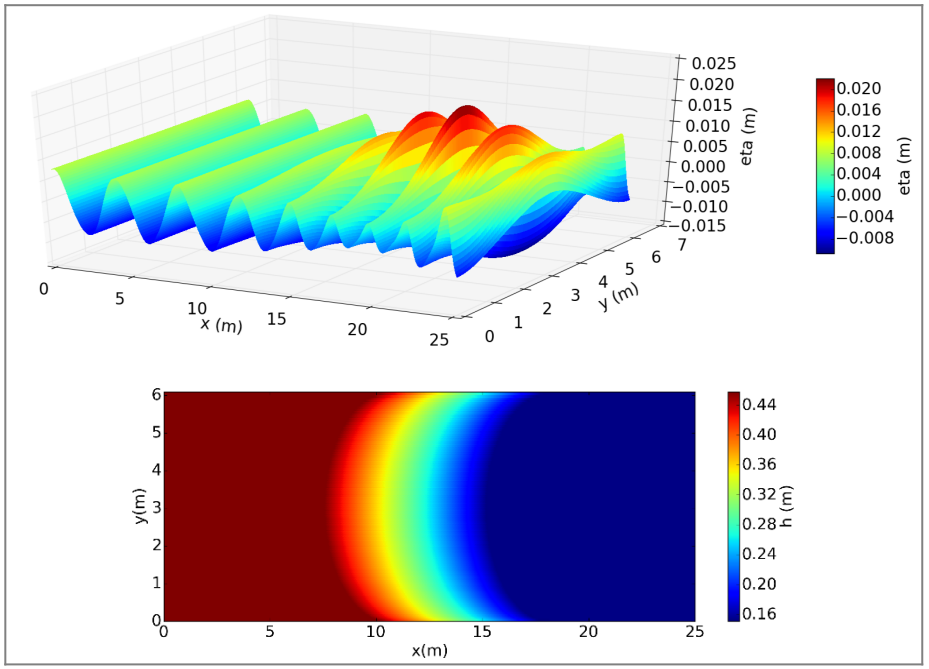


Deep ocean → **Shallow water**

Wave transformation processes

II. Numerical wave modeling

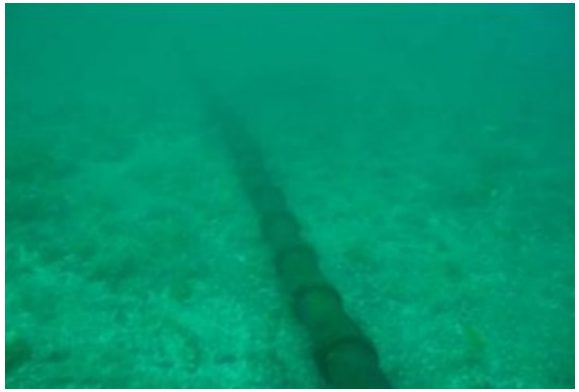


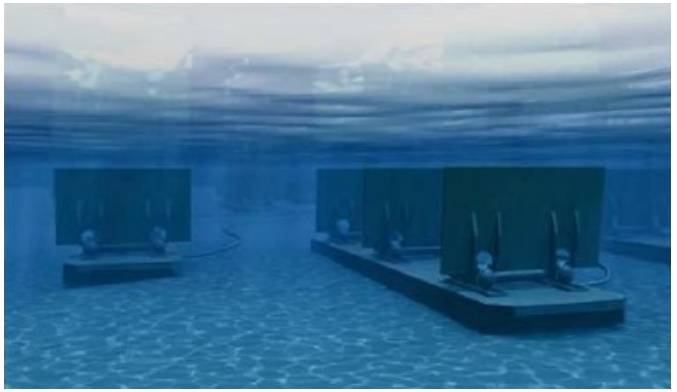


Phase-averaged (spectral) wave models

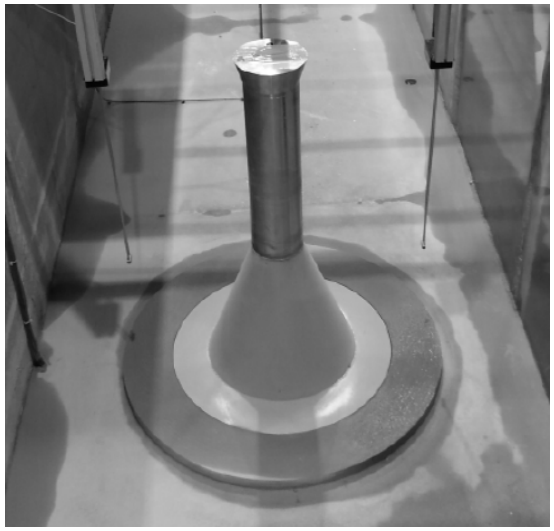
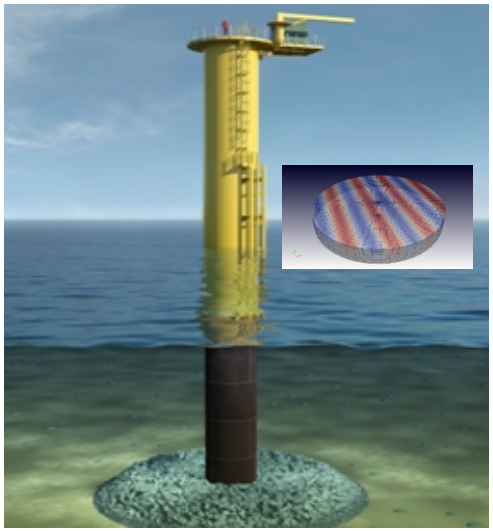
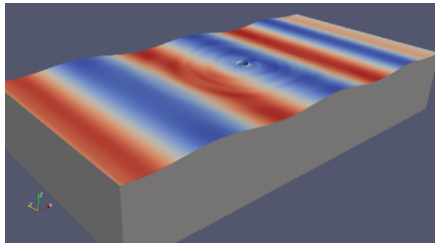
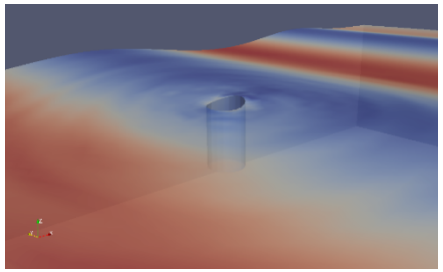
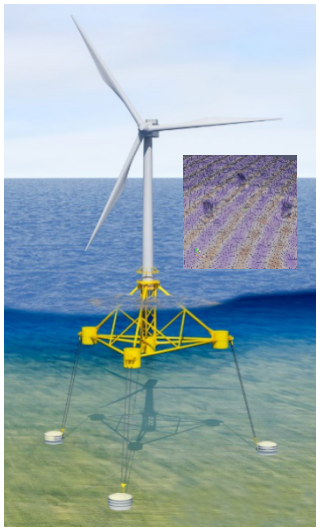
Phase-resolving (deterministic) wave models

III. Wave-structure interactions





For a variety of different applications…

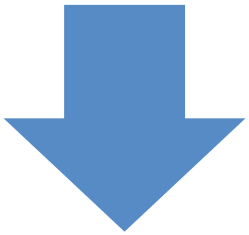






Going from laboratory experiments…
to numerical modeling… to prototypes

Offshore wind farm project

Characterization, modeling and design of a wind farm

- **Applying theories presented in class**
- **Wave data analysis**
- **Running simple models and comparing results**
- **Modeling wind turbine stability**



Comprehension of course material *(group analyses)*

Synthesizing results *(individual reports)*

Clear presentation of your work *(final oral presentation)*

Project schedule

Due dates in red: reports (R) and final presentation (FP)

I. Wave dynamics	Project	Lecturer	Date	
1. Introduction and characterizing ocean waves and sea states	Description of class project and site selection	M. Yates	25/09	
2. Linear wave theory	Characterizing offshore waves	M. Yates	09/10	
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II. Numerical wave modeling				
5. Numerical wave modeling I	Introduction to wave propagation model	J. Harris	**23/10****	**R1**
6. Numerical wave modeling II	Estimating mean and extreme nearshore waves	J. Harris	13/11	
III. Wave-structure interactions				
7. Dynamics of a body in waves	Wave load estimation	J. Harris	20/11	
8. Modeling wave-body interactions	Calculating movement of floating wind turbine	C. Peyrard	**04/12**	**R2**
9. Offshore wind turbine foundations	Final project work session	C. Peyrard	11/12	
10. Final project presentations 14h - 18h15			***18/12***	**FP**

Student mini-presentation topics (class 4)

Objective: present to the class a subject that interests you

Presentation format :

- **Work in pairs (2-3)**
- **10 minutes to present:**
  - **Introduction/context**
  - **Description of subject, presentation of important concept**
  - **Conclusion/take home message**
- **3-5 minutes for questions**

Student mini-presentation topics (class 4)

1. Wave buoy and pressure sensor measurements
2. Remote sensing (satellite and radar) wave measurements
3. Nonlinear wave theories (Stokes 2nd order, …)
4. Stokes drift (difference between Lagrangian and Eulerian averages of a flow field)
5. Rip currents
6. Infragravity waves or « surf beat »
7. Effects of bottom friction
8. Surf zone drifter studies
9. Physics of wave surfing
10. Wave energy converters
11. Tidal wave energy
12. Internal waves
13. Internship projects that you've worked on
14. Other topics of interest?

Student mini-presentation topics (class 4)

1 Wave buoy and pressure sensor measurements

2 **Subjects presented in 2025**

3 Meteotsunamis

4 Tidal wave energy

5 Wave energy converters

6 Wave buoy and pressure sensor measurements

7 Remote sensing (satellite and radar) wave measurements

8 Internal waves

9 Physics of surfing

1 Rip currents

1 Wave-current interactions

1 Impacts of waves on coasts

1

14 Other topics of interest?

Student mini-presentation topics (class 4)

Presentations on November 6, 2026

SIGN UP (class 2) for your presentations by 09/10/2026: please indicate the topic and the group members (2-3 people/group).

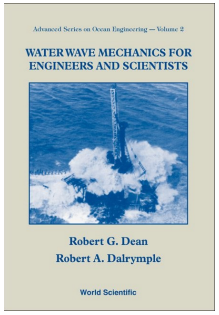
https://lite.framacalc.org/d68gl2xpn6-anow

Class materials

1. **Class notes + presentations available here:**

   https://cloud.enpc.fr/s/PcBgoKQsD2T2rPH

1. **Reference textbooks for more information:**



R. G. Dean and R. Dalrymple. Water Wave Mechanics for Engineers and Scientists. World Scientific Publishing, 1991.

Y. Goda. Random Seas and Design of Maritime Structures. World Scientific Publishing, 2000.

L. H. Holthuijsen. Waves in Oceanic and Coastal Waters. Cambridge University Press, 2007.

P. K. Kundu. Fluid mechanics. Academic Press, London, sixth edition. edition, 2016.

C. C. Mei. The Applied Dynamics of Ocean Surface Waves. World Scientific Publishing, 1992.

USACE. Shore Protection Manual. Technical report, U.S. Army Corps of Engineers, Vicksburg, Mississippi, 1984.

Bring a laptop (if available) to work on your project